

NJS_Q08_M: Sync Offloading / Worker Pool

Difficulty: MEDIUM/HARD

Scenario

A Node.js web application is exhibiting runtime anomalies or configuration errors in production. You are tasked with writing clean, async, and correct JavaScript code to solve the requirements.

Username & Naming Conventions

- Resources must utilize your candidate suffix prefix (<username>-<exam_code>).

Task Objectives

- Node.js Optimization & Code Resolution:** Offload a CPU-bound hashing workload from the main event loop using `worker_threads`.
- Ensure your changes pass all local verification tests.

Verification

- Local run test script `run.py` can be executed to verify local workspace properties.
- Central validation runs queries against the live workspace to verify integration.

Grading Criteria

- TC1: File `server.js` exists in workspace** (4 Marks)
- TC2: File `worker.js` exists in workspace** (4 Marks)
- TC3: Worker pools utilized for processing CPU-intensive operations** (4 Marks)
- TC4: Main execution threads remain non-blocked during hashing execution** (4 Marks)
- TC5: Asynchronous response payloads correctly formatted** (4 Marks)

Important Notes

- Always check that your syntax compiles without errors.